

Overview of the Neutrinoless Double Beta Decay

and the Majorana Nature of the Neutrino

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Abstract

Neutrino oscillations have established that neutrinos are massive, but they cannot determine whether the mass term is of Dirac or of Majorana type, that is, whether the total lepton number is conserved. Settling this question requires a process violating lepton number by two units, and neutrinoless double beta decay is the only one accessible with present technology. This review presents the field-theoretical distinction between the two hypotheses, the Weinberg operator and the seesaw mechanism, and the proof that oscillations are insensitive to the Majorana phases. It then develops the phenomenology of the decay, from the nuclear structure that makes it observable to the effective Majorana mass and the nuclear matrix elements connecting it to a measured half-life, where the quenching of the axial coupling g_A is identified as the dominant theoretical uncertainty. Experimental strategies, backgrounds and current limits are reviewed, the most stringent being $T_{1/2}^{0\nu}(^{136}\text{Xe}) > 3.8 \times 10^{26}$ yr from KamLAND-Zen, and the prospects of the next generation are discussed in the light of cosmological constraints and of alternative lepton-number-violating mechanisms.

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1 Introduction

The discovery of neutrino oscillations closed one question in particle physics and opened another. Atmospheric data from Super-Kamiokande [1] and solar data from SNO [2, 3] established that neutrinos change flavour as they propagate, which requires at least two mass eigenstates to be non-degenerate. Massless neutrinos, as assumed in the minimal Standard Model, are therefore excluded by experiment.

Oscillations answer only part of the question. They probe the squared mass differences $\Delta m_{ij}^2 = m_i^2 - m_j^2$, not the absolute masses, and carry no information on the mechanism generating them. A massive fermion field admits a mass term in two structurally different ways. A *Dirac* mass couples a left-handed to a right-handed field and conserves the total lepton number L , exactly as for the electron. A *Majorana* mass, possible only for an electrically neutral field, identifies the particle with its own antiparticle and violates L by two units. Which is realised in nature remains unknown, and oscillation probabilities are provably insensitive to the additional CP-violating phases that distinguish the two cases [4]. Settling the question requires an observable sensitive to $|\Delta L| = 2$ rather than $|\Delta L| = 0$.

Among all conceivable lepton-number-violating processes, only one is experimentally accessible today, neutrinoless double beta decay,

$$(A, Z) \rightarrow (A, Z + 2) + 2e^-, \quad 0\nu\beta\beta, \quad (1)$$

in which two neutrons convert into two protons with only two electrons emitted and no neutrinos in the final state [5].

Observing $0\nu\beta\beta$ would establish that the total lepton number is not conserved in nature, a discovery of the same conceptual weight as proton decay for the baryon number. Under general assumptions, the Schechter–Valle theorem [6] further guarantees that such an observation implies a Majorana mass term for the electron neutrino, independently of which new-physics mechanism triggers the decay.

This thesis is organised as follows. Section 1 introduces the distinction between Dirac and Majorana neutrinos, the Weinberg operator and the seesaw mechanism, and the insensitivity of oscillations to the Majorana phases. Section 2 develops the phenomenology of double beta decay, from the nuclear physics making the process observable to the effective Majorana mass $m_{\beta\beta}$ and the nuclear matrix elements connecting it to a measured half-life. Section 3 reviews experimental strategies, backgrounds and current limits, and Section 4 places $0\nu\beta\beta$ in the broader context of cosmology and alternative mechanisms, before Section 5 summarises the present status of the field.

1.1 Charge conjugation and the reality condition

For a spinor field ψ , the charge-conjugate field is defined by

$$\psi^c \equiv C\bar{\psi}^t, \quad \bar{\psi} \equiv \psi^\dagger \gamma^0, \quad (2)$$

where C is the charge-conjugation matrix, satisfying $C^{-1}\gamma_\alpha C = -\gamma_\alpha^t$. Physically, ψ^c describes the antiparticle associated with ψ , every additive quantum number changes sign under this operation, while mass and spin are unchanged. For a Dirac field, ψ and ψ^c are independent and describe distinct particles. Majorana's 1937 proposal [7] is that a field may instead satisfy a reality condition, identifying it with its own charge conjugate [4].

A field χ is a Majorana field if it satisfies

$$\chi = C\bar{\chi}^t \iff \chi = \chi^c, \quad (3)$$

so that the particle it describes is identical to its own antiparticle.

This is possible only for a field carrying no conserved additive quantum number, a charge Q would have to satisfy $Q = -Q$. Among the elementary fermions, the neutrino is the only electrically neutral one, and hence the only viable candidate. Whether it does satisfy the reality condition, that is, whether the lepton number L it might otherwise carry is conserved, is the question this thesis addresses.

1.2 Degrees of freedom and mass terms

The reality condition ties two of the four spinor components to the other two, halving the number of independent degrees of freedom. In the rest frame, a Dirac field therefore describes four states, particle and antiparticle with either spin orientation, whereas a Majorana field describes only two, the two spin states of a single self-conjugate particle. Its free Lagrangian consequently carries a factor one half, $\mathcal{L} = \frac{1}{2}\bar{\chi}(i\not{\partial} - m)\chi$, which compensates the reduced field content.

The construction is not merely formal [4]. From the single left-handed field ν_L already present in the Standard Model, and without introducing any new particle content, one may build

$$\chi \equiv \nu_L + C\bar{\nu}_L^t, \quad (4)$$

which satisfies $\chi = \chi^c$, the original field being recovered by projection, $\nu_L = P_L\chi$. The Majorana hypothesis is in this sense the more economical one, a Dirac neutrino instead requires three additional right-handed fields ν_R carrying no Standard Model gauge charge, and therefore sterile.

The distinction becomes sharper for the mass term. A Dirac mass, $\mathcal{L}_D = -m_D \bar{\nu}_R \nu_L + \text{h.c.}$, couples the left-handed field to an independently introduced right-handed partner and conserves

L . A Majorana mass instead uses the same field twice,

$$\mathcal{L}_M = \frac{1}{2} \nu_{\ell L}^t C^{-1} M_{\ell\ell} \nu_{\ell L} + \text{h.c.}, \quad (5)$$

and violates L by two units. Gribov and Pontecorvo established two structural consequences of the anticommuting nature of fermionic fields [8, 9]. The flavour matrix M is necessarily symmetric, $M^t = M$, only its symmetric part surviving the antisymmetry of C^{-1} . And its diagonalisation, through the Takagi decomposition $M = U^t \text{diag}(m_1, m_2, m_3) U$ for complex symmetric matrices rather than the Hermitian eigenvalue problem, always yields mass eigenstates that are themselves Majorana particles.

1.3 The Weinberg operator and the seesaw mechanism

The Standard Model conserves B and L , but only accidentally, no operator of dimension four or less compatible with its gauge symmetry violates either. Non-perturbative sphaleron processes conserve $B - L$ alone, which suffices to forbid $0\nu\beta\beta$ entirely. An exact global symmetry of this kind is nonetheless regarded with suspicion, as emphasised by Witten, “in a modern understanding of particle physics, global symmetries are approximate” [9, 10]. The symmetry $U(1)_{LC}$ responsible for $L = \text{const.}$, and hence for purely Dirac neutrinos, is expected to be broken, if only by quantum gravity effects.

In the language of effective field theory, exactly one operator of dimension five can be built from the lepton doublet ℓ_L and the Higgs doublet H ,

$$\mathcal{O}_5 = \frac{(\ell_L H)(\ell_L H)}{\Lambda}. \quad (6)$$

After electroweak symmetry breaking, $\langle H \rangle = v$, it generates a Majorana mass $m_\nu \sim v^2/\Lambda$ and violates L [11]. It is the lowest-dimension, hence least suppressed, operator beyond the Standard Model, and it is Majorana by construction.

With $v \simeq 174$ GeV and $m_\nu \sim 0.05$ eV this gives $\Lambda \sim 10^{14}$ to 10^{15} GeV, close to the grand unification scale, so that the smallness of neutrino masses becomes evidence for a high new-physics scale rather than an anomaly.

The type I seesaw is the best known ultraviolet completion of $\mathcal{O}_5$. Adding right-handed neutrinos N_R , singlets under every gauge group, nothing forbids them a Majorana mass M unrelated to the electroweak scale. In the one-flavour case the mass matrix in the basis (ν_L, N_R) reads

$$\begin{pmatrix} 0 & m_D \\ m_D & M \end{pmatrix}, \quad (7)$$

whose diagonalisation for $M \gg m_D$ yields one heavy eigenstate of mass $\simeq M$ and one light eigenstate of mass

$$m_\nu \simeq -\frac{m_D^2}{M}, \quad (8)$$

the heavier M , the lighter m_ν , whence the name. With m_D of order the top quark mass and $M \sim 10^{15}$ GeV one obtains $m_\nu \sim 10^{-2}$ eV without any small coupling put in by hand. Bilenky and Pontecorvo generalised this to the most general Dirac plus Majorana mass term built from n active and n sterile fields, whose diagonalisation yields $2n$ massive Majorana neutrinos [9, 12].

1.4 Majorana phases and the blindness of oscillations

The number of physical CP-violating phases in the mixing matrix U_{PMNS} depends on the nature of the neutrino. A Dirac mass term $m_j \bar{\nu}_{jR} \nu_{jL}$ is invariant under an independent rephasing of each mass eigenstate, provided the right-handed partner is rephased identically, and this freedom removes phases from the mixing matrix. A Majorana mass term has no right-handed partner to compensate the rephasing, which therefore alters the mass term itself, only the discrete freedom $\nu_j \rightarrow -\nu_j$ survives. Majorana neutrinos thus carry additional, irremovable phases with no Dirac counterpart.

For n families there are $(n-1)(n-2)/2$ Dirac phases in both cases, plus $(n-1)$ Majorana phases if the neutrino is self-conjugate [4, 13]. For $n = 3$, one Dirac phase δ and two Majorana phases α_{21} , α_{31} , and

$$U_{\text{PMNS}} = V \cdot P, \quad P = \text{diag}\left(1, e^{i\alpha_{21}/2}, e^{i\alpha_{31}/2}\right), \quad (9)$$

with V the usual matrix of three angles and one phase.

Even two mixed Majorana neutrinos retain one physical CP-violating phase, whereas the mixing of two Dirac neutrinos, or of two quarks, is automatically CP-conserving.

Oscillation probabilities in vacuum and in matter do not depend on the Majorana phases [13, 14]. They are built from combinations $U_{\ell i} U_{\ell' i}^* U_{\ell j}^* U_{\ell' j}$ in which P enters through $PP^\dagger = \mathbb{1}$ and cancels identically. No oscillation experiment, of any precision, can therefore probe the nature of the neutrino.

This motivates the search for an observable sensitive to a genuine $|\Delta L| = 2$ transition, whose effective parameter does depend on the Majorana phases, and is constructed in Section 2.4.

2 Phenomenology of double beta decay

2.1 Nuclear stability and the choice of candidate isotopes

Before considering the role of the neutrino, double beta decay is first a nuclear physics problem, its existence as an observable process follows from nuclear structure rather than from particle physics.

The stability of a nucleus (A, Z) against beta decay is governed by its binding energy,

captured by the semi-empirical mass formula [15],

$$M(A, Z) = Zm_p + Nm_n - a_V A + a_S A^{2/3} + a_C \frac{Z^2}{A^{1/3}} + a_A \frac{(N - Z)^2}{A} + \delta(A, Z), \quad (10)$$

whose last term, the pairing energy, is negative for even-even nuclei, positive for odd-odd nuclei and zero for odd- A nuclei. At fixed even A the isobars therefore split into two parabolas in the plane $M(A, Z)$ against Z , the even-even one lying lower.

Because of the pairing term, an even-even nucleus can be simultaneously heavier than the isobar $(A, Z + 2)$ and lighter than the intermediate odd-odd isobar $(A, Z + 1)$. Ordinary beta decay is then energetically forbidden and the only open channel is the second-order transition directly to $(A, Z + 2)$, double beta decay.

All candidates are consequently even-even, with ground state spin 0^+ . Four channels are in principle allowed, $\beta^-\beta^-$, $\beta^+\beta^+$, double electron capture and $\text{EC}\beta^+$, the last three having a Q -value reduced by $2m_e c^2$ or $4m_e c^2$ and being correspondingly harder to observe. This thesis focuses on $\beta^-\beta^-$, for which $Q_{\beta\beta} = M(A, Z) - M(A, Z + 2)$ in atomic masses.

The choice among the isotopes of Table 1 balances three requirements. A high $Q_{\beta\beta}$ is the dominant criterion, natural radioactivity ending at the 2615 keV line of ^{208}Tl from the ^{232}Th chain, so that pushing $Q_{\beta\beta}$ above this value removes the main background. High isotopic abundance, or compatibility with efficient enrichment, gas centrifugation requiring a stable hexafluoride compound, reduces cost and increases the achievable source mass. Finally the isotope must suit a scalable detection technology, discussed in Section 3.

Table 1: Natural isotopic abundance and $Q_{\beta\beta}$ value for the most commonly studied double-beta emitters [15].

Isotope	Abundance (%)	$Q_{\beta\beta}$ (MeV)
^{48}Ca	0.187	4.263
^{76}Ge	7.8	2.039
^{82}Se	9.2	2.998
^{96}Zr	2.8	3.348
^{100}Mo	9.6	3.035
^{116}Cd	7.6	2.813
^{130}Te	34.08	2.527
^{136}Xe	8.9	2.459
^{150}Nd	5.6	3.371

^{76}Ge shows that no single criterion dominates. Its $Q_{\beta\beta}$ lies below the ^{208}Tl line, yet it remains among the most competitive isotopes because high-purity germanium detectors combine exceptional energy resolution with remarkable intrinsic radiopurity. A further empirical fact is that the phase-space factor and the nuclear matrix element are systematically anticorrelated across candidate nuclei [4, 16], so that the expected rate per unit mass is nearly uniform and no isotope is intrinsically favoured by nuclear structure alone.

2.2 Two-neutrino double beta decay

The two-neutrino mode,

$$(A, Z) \rightarrow (A, Z + 2) + 2e^- + 2\bar{\nu}_e, \quad 2\nu\beta\beta, \quad (11)$$

was first proposed by Goeppert-Mayer in 1935, who estimated its half-life to exceed 10^{17} years [17]. It is allowed in the Standard Model, but only as a second-order weak process, hence its extraordinarily long lifetime. It has since been observed in twelve nuclei, with half-lives from 10^{18} to 10^{24} years [15].

The rate factorises into kinematic and nuclear parts,

$$\Gamma^{2\nu} = \frac{1}{T_{1/2}^{2\nu}} = G^{2\nu}(Q_{\beta\beta}, Z) |M^{2\nu}|^2, \quad (12)$$

where the phase-space factor $G^{2\nu}$ can be computed exactly, while the nuclear matrix element $M^{2\nu}$ is far harder to evaluate.

The phase-space factor integrates over the momenta of the four emitted leptons, weighted by the Fermi function accounting for the Coulomb distortion of the outgoing electrons, the explicit triple integral is given in Section B [15]. It yields a remarkably strong dependence on the Q -value, $G^{2\nu} \propto Q_{\beta\beta}^{11}$, against the much milder $G^{0\nu} \propto Q_{\beta\beta}^5$ of the neutrinoless mode derived in Section 2.3. The difference in exponent has a simple origin, four leptons share the available energy in the two-neutrino final state against only two in the neutrinoless one, each additional particle contributing further powers of $Q_{\beta\beta}$. This is also why Furry's original estimate, made before the $V-A$ structure of the weak interaction was established, wrongly anticipated that $0\nu\beta\beta$ would necessarily be faster than $2\nu\beta\beta$.

The transition connects two 0^+ nuclei through virtual 1^+ states of the intermediate odd-odd nucleus, and isospin conservation forbids the Fermi part of the operator, so that only the Gamow-Teller contribution survives [15],

$$M_{\text{GT}}^{2\nu} = \sum_m \frac{\langle 0_f^+ | \tau^+ \sigma | 1_m^+ \rangle \langle 1_m^+ | \tau^+ \sigma | 0_i^+ \rangle}{E_m - (M_i + M_f)/2}, \quad (13)$$

the numerator being the product of the amplitudes of the two successive beta decays and the

denominator the energy of the virtual intermediate state.

The two-neutrino mode plays a dual role for the neutrinoless search. Since $M^{2\nu}$ follows directly from the measured half-life, it provides the only direct experimental test of the nuclear models used to predict $M^{0\nu}$, as discussed in Section 2.6. Conversely, its continuous spectrum extends with a falling tail all the way to $Q_{\beta\beta}$, and constitutes an irreducible background to $0\nu\beta\beta$, produced by the very isotope under study and impossible to shield against. The first observation was geochemical, Inghram and Reynolds inferring a ^{130}Te half-life of 1.4×10^{21} years from an excess of ^{130}Xe in an ancient ore [18], and it took nearly forty years before a direct counter measurement, on ^{82}Se with only 36 candidate events [19].

2.3 The neutrinoless mechanism

Proposed by Furry in 1939 [5], the neutrinoless channel proceeds through the exchange of a single virtual neutrino between two neutrons undergoing beta decay inside the same nucleus, rather than through two independent decays each releasing its own antineutrino. Two conditions, both necessary, follow directly from the field-theoretical structure established in Section 1.

- (i) The neutrino emitted at the first vertex must be absorbable at the second, which requires $\nu = \bar{\nu}$, a Majorana neutrino. (ii) The charged current couples only left-handed chirality, so the state emitted with positive helicity at the first vertex must be reabsorbed with negative helicity at the second, a flip that is only possible because the neutrino is massive, chirality and helicity coincide exactly only for a massless particle. The amplitude is therefore proportional to m_ν , not merely allowed by it.

This second point admits a compact derivation. The Majorana condition of Section 1.1 allows a propagator absent in the Dirac theory, $\langle 0|T[\chi(x)\chi(y)]|0\rangle = -\Delta(x-y)C$. Sandwiched between the left-chiral projectors of the two weak vertices, the ordinary Dirac propagator reduces to

$$P_L \Delta(x) P_L = P_L \int \frac{d^4 q}{(2\pi)^4} \frac{i m}{q^2 - m^2 + i0} e^{-iqx}, \quad (14)$$

the term proportional to $\not{q}$, which would survive for a massless neutrino, is annihilated by the chiral projectors, $P_L \not{q} P_L = 0$, and only the mass term remains.

The virtual neutrino carries an impulse set by the nucleon spacing inside the nucleus, $|\vec{q}| \sim \hbar c / 1 \text{ fm} \sim 100 \text{ MeV}$. Comparing this scale to the mass of the exchanged state separates two regimes. For a light neutrino, $m \ll |\vec{q}|$, the propagator gives m/q^2 and the rate is governed by the coherent sum $m_{\beta\beta} \equiv |\sum_j U_{ej}^2 m_j|$, constructed in Section 2.4. For a heavy neutrino, $m \gg |\vec{q}|$, the interaction becomes point-like and the rate is instead suppressed as $1/M$, a case revisited in Section 4.2. Restricting to the light-neutrino mechanism, the half-life factorises as

$$\left[T_{1/2}^{0\nu}\right]^{-1} = G^{0\nu}(Q_{\beta\beta}, Z) |M^{0\nu}|^2 \left(\frac{m_{\beta\beta}}{m_e}\right)^2, \quad (15)$$

the same three-way split, kinematics, nuclear structure, particle physics, already met for $2\nu\beta\beta$ in Section 2.2.

Because the nuclear recoil is negligible, the full transition energy goes into the two electrons, $T_{e1} + T_{e2} = Q_{\beta\beta}$, so the summed-energy spectrum is a single line at $Q_{\beta\beta}$, broadened only by the detector resolution, rather than the continuum of Section 2.2. This exceptionally clean signature, developed experimentally in Section 3.1, is what makes the search feasible despite the extreme rarity of the process.

2.4 The effective Majorana mass $m_{\beta\beta}$

In the flavour basis, the only element of the Majorana mass matrix that violates the electronic lepton number by two units is M_{ee} , which is therefore the quantity governing the $0\nu\beta\beta$ amplitude [4]. Written in the mass basis, it defines the effective Majorana mass.

$$m_{\beta\beta} \equiv \left| \sum_{j=1}^3 U_{ej}^2 m_j \right| = \left| m_1 c_{12}^2 c_{13}^2 + m_2 s_{12}^2 c_{13}^2 e^{i\alpha_{21}} + m_3 s_{13}^2 e^{i(\alpha_{31}-2\delta)} \right|, \quad (16)$$

with $s_{ij} = \sin \theta_{ij}$, $c_{ij} = \cos \theta_{ij}$, and $\Gamma^{0\nu} \propto m_{\beta\beta}^2$ for the light-neutrino mechanism.

Two features of this expression deserve emphasis. First, it involves the *squares* U_{ej}^2 , not the moduli $|U_{ej}|^2$, so the Majorana phases introduced in Section 1.4 survive and the three terms may interfere destructively. Second, $m_{\beta\beta}$ is a coherent sum, and is therefore distinct from the two other observables sensitive to the absolute mass scale, the single beta decay parameter $m_\beta^2 = \sum_j |U_{ej}|^2 m_j^2$, probed by spectrometers such as KATRIN, and the cosmological sum $\Sigma = \sum_j m_j$. Of the three, only $m_{\beta\beta}$ presupposes that the neutrino is a Majorana particle.

Allowing the phases to vary freely, the extremal values of $m_{\beta\beta}$ are

$$m_{\beta\beta}^{\max} = \sum_j |U_{ej}^2| m_j, \quad m_{\beta\beta}^{\min} = \max \left\{ 2|U_{ej}^2| m_j - m_{\beta\beta}^{\max}, 0 \right\}, \quad (17)$$

derived in Section C. The consequence is important, there exist regions of parameter space in which $m_{\beta\beta}$ vanishes exactly even though the neutrino is a Majorana fermion, so that a non-observation of $0\nu\beta\beta$ can never establish the Dirac nature of the neutrino.

Since the sign of Δm_{31}^2 is not determined experimentally, two mass orderings remain possible, and the expected value of $m_{\beta\beta}$ depends on both the ordering and the mass of the lightest neutrino, as summarised in Table 2.

Table 2: Neutrino mass spectra and the corresponding expectations for $m_{\beta\beta}$ [4, 9].

Spectrum	Ordering	Condition	Expected $m_{\beta\beta}$
Normal hierarchical (NH)	$m_1 \ll m_2 < m_3$	$m_1 \rightarrow 0$	1 to 5 meV, may vanish
Inverted hierarchical (IH)	$m_3 \ll m_1 < m_2$	$m_3 \rightarrow 0$	15 to 50 meV
Quasi-degenerate (QD)	$m_1 \simeq m_2 \simeq m_3$	$m \gtrsim 0.1$ eV	$\gtrsim 50$ meV

The inverted band does not extend below roughly 15 meV, which sets the quantitative target of the next generation of experiments discussed in Section 4, a sensitivity of order 10 meV would explore the inverted scenario in full, either yielding a signal or excluding it.

2.5 The Schechter–Valle theorem

The preceding derivation shows that a light Majorana neutrino produces $0\nu\beta\beta$. The converse question is more delicate, if the decay were triggered by some other mechanism entirely, could anything still be concluded about the nature of the neutrino?

Schechter and Valle showed in 1982 that, under very general assumptions, an observation of $0\nu\beta\beta$ implies the existence of a Majorana mass term for ν_e , and hence massive Majorana neutrinos, irrespective of the mechanism triggering the decay [6].

The argument proceeds in four steps [9]. An observed $0\nu\beta\beta$ implies an elementary process in which the electron, up-quark and down-quark fields each appear twice, whatever its internal content, enclosed in a black box resembling the dimension-nine effective operator. Using W bosons, the two quark pairs can be contracted. The electron and W fields can then be converted into neutrino fields, producing a contribution to the Majorana neutrino mass. Finally, the possibility that this contribution be exactly cancelled by others is excluded as unnatural.

A quantitative evaluation shows that the black-box diagram generates the mass only at four-loop order, of order $\delta m_\nu \sim 10^{-24}$ to 10^{-28} eV, many orders of magnitude below the scale implied by oscillations [4, 9]. The qualitative statement stands, but the diagram itself explains nothing of the observed masses, other operators must dominate. The naturality step is moreover not rigorously defined, and models exist in which $0\nu\beta\beta$ occurs while the induced Majorana mass vanishes. The result is best read as a robust reminder that any theory containing Majorana masses links them to $0\nu\beta\beta$, rather than as a theorem in the strict sense.

This theorem has nonetheless provided, since the 1980s, the central theoretical motivation for pursuing experimental searches with ever increasing sensitivity, whatever the underlying new physics, a signal would mean Majorana.

2.6 Nuclear matrix elements and the quenching of g_A

Inverting the factorised rate of Section 2.3, an experimental limit on the half-life translates into a limit on the effective Majorana mass,

$$m_{\beta\beta} \leq \frac{m_e}{|\mathcal{M}| \sqrt{G^{0\nu} T_{1/2}^{0\nu}}}, \quad (18)$$

so that extracting particle physics from a measurement requires knowing both $G^{0\nu}$ and $\mathcal{M}$.

The phase-space factors are under good control. Modern evaluations solve numerically for the Dirac wave functions and include relativistic corrections, finite nuclear size and atomic screening, leaving a residual uncertainty of about 7 % [4]. The nuclear matrix elements are the difficulty. Computing $\mathcal{M}$ requires describing the wave functions of two open-shell nuclei of

several tens of nucleons together with the operator connecting them, a task addressed by several methods with opposite philosophies, the interacting shell model, which retains all correlations within a limited valence space, the quasiparticle random phase approximation, which uses a large valence space but only certain correlations, and the interacting boson model, which describes low-lying states in terms of s and d bosons. Agreement between methods has improved considerably, from a spread exceeding a factor four in the 1984 to 1998 calculations down to a factor two to three today, each scheme now quoting an intrinsic uncertainty of order 15 to 20 % [4]. It is the discrepancy between schemes, not the internal error of any one of them, that should be taken as the overall uncertainty.

A further and larger source of uncertainty concerns the axial coupling constant. Writing $\mathcal{M} = g_A^2 (M_{\text{GT}}^{(0\nu)} - (g_V/g_A)^2 M_F^{(0\nu)} + M_T^{(0\nu)})$, the free-nucleon value $g_A \simeq 1.269$ is known to be renormalised, or quenched, by the nuclear medium, a problem first identified when attempting to reproduce measured $2\nu\beta\beta$ rates. The proposed causes include the truncation of the model space, contributions from non-nucleonic degrees of freedom, and the renormalisation of the Gamow-Teller operator by two-body currents [4].

Since $\mathcal{M} \propto g_A^2$, one has $T_{1/2}^{0\nu} \propto (g_A^{\text{eff}})^{-4}$. Reducing g_A by a factor $(1 - \delta)$ suppresses the signal as $(1 - \delta)^4$, and recovering the same statistical significance in the presence of background requires an exposure larger by $1/(1 - \delta)^8$, a factor 2.3 for a 10 % quenching and 6 for 20 %. Depending on whether the free-nucleon, quark or strongly quenched value is adopted, the limit inferred on $m_{\beta\beta}$ from a given half-life varies by a factor four to eight [4], by far the dominant theoretical uncertainty in the field.

Whether the quenching is identical in the two modes remains open. The dominant multipole in $0\nu\beta\beta$ is also of Gamow-Teller type, which argues for a common treatment, but $2\nu\beta\beta$ proceeds exclusively through 1^+ intermediate states whereas $0\nu\beta\beta$ passes through all multipoles, and transitions through other spin-parity states might be unquenched. Measuring $2\nu\beta\beta$ half-lives therefore constrains, without fully determining, the calculations relevant to the neutrinoless mode.

3 Experimental search for $0\nu\beta\beta$

3.1 Signature and sensitivity

The experimental signature is the monochromatic line at $Q_{\beta\beta}$ established in Section 2.3, sitting on the falling tail of the $2\nu\beta\beta$ continuum. Since the two modes differ only by the presence of the line, the ratio of counts in the region of interest scales as

$$R_{0\nu/2\nu} \propto \left(\frac{Q_{\beta\beta}}{\Delta} \right)^6 \frac{T_{1/2}^{2\nu}}{T_{1/2}^{0\nu}}, \quad (19)$$

with Δ the full width at half maximum [4]. The sixth power explains why energy resolution is the single most valuable detector property.

In the absence of a signal, the sensitivity is conventionally expressed as the half-life corresponding to the largest signal that background fluctuations could conceal,

$$S^{0\nu} \propto \varepsilon \frac{x \eta N_A}{\mathcal{M}_A} \sqrt{\frac{M \cdot t}{B \cdot \Delta}}, \quad (20)$$

where ε is the detection efficiency, η the isotopic abundance, M the detector mass, t the measuring time and B the background index per unit mass, energy and time [4].

When the background is so low that the expected number of counts in the region of interest over the whole experiment is of order unity, $M \cdot t \cdot B \cdot \Delta \lesssim 1$, the experiment enters the zero-background regime and the sensitivity ceases to grow as $\sqrt{Mt}$, improving instead linearly with $M \cdot t$. This is the regime the next generation aims for, and it inverts the usual strategy, a cleaner detector becomes more valuable than a larger one.

3.2 Backgrounds

Four categories dominate [4, 15]. Natural α , β and γ radioactivity from traces of ^{238}U and ^{232}Th in every material imposes radiopurity requirements below 0.1 ppb in fiducial volumes, degraded surface α particles producing a continuum extending into the region of interest. Radon, a noble gas present in both chains, diffuses from rock and materials, its daughters ^{214}Bi and ^{208}Tl being particularly dangerous because of their high Q -values, and is mitigated by delayed β - α coincidence tagging and by flushing with radon-free gas. Cosmic muons, whose flux at sea level would overwhelm any detector, require an underground site, a laboratory such as Gran Sasso providing a reduction of about 10^6 . Neutrons from spontaneous fission and (α, n) reactions contribute a flat spectrum through radiative capture, and therefore matter more for $0\nu\beta\beta$ than for $2\nu\beta\beta$. Shielding is correspondingly layered, lead or iron against external γ rays, radiopure copper inside it, and a hydrogen-rich doped material such as borated polyethylene to thermalise and capture neutrons.

To these instrumental backgrounds must be added the irreducible $2\nu\beta\beta$ tail of Section 2.2, which no shielding can suppress since it originates in the isotope itself.

3.3 Detector strategies

Two families of detectors coexist, differing in whether source and detector coincide [15].

In the *calorimetric* approach the isotope constitutes the detector itself, as in high-purity germanium diodes, TeO_2 bolometers, liquid xenon time projection chambers or xenon-loaded liquid scintillators. Large masses and high efficiency follow naturally, and the energy resolution can be excellent, of order 0.1 % FWHM for germanium and bolometers. The drawback is that the readout gives energy alone, so topological reconstruction is difficult except in time projection chambers, at the cost of resolution.

In the *external source* approach the isotope is a thin foil surrounded by a tracker and a calorimeter, allowing the two electron tracks, their common vertex and their relative angle to

be reconstructed, as in NEMO-3 and SuperNEMO. Event-by-event background rejection is correspondingly powerful, but the resolution is poor, around 10 %, limited by energy losses in the foil, the efficiency is of order 30 %, and masses beyond about 100 kg are difficult.

The best limits on $0\nu\beta\beta$ consequently come from calorimeters, while tracking detectors have produced the most detailed $2\nu\beta\beta$ measurements. Should a signal be found, topological reconstruction would become decisive again in identifying the underlying mechanism, as discussed in Section 4.2.

3.4 Current limits

No signal has been observed. The most stringent constraints, collected in Table 3, come from three isotopes, the ranges quoted on $m_{\beta\beta}$ reflecting the nuclear matrix element uncertainties of Section 2.6 rather than any experimental error.

Table 3: Best current limits on $0\nu\beta\beta$ half-lives at 90 % C.L. and the corresponding effective Majorana mass ranges.

Isotope	Experiment	$T_{1/2}^{0\nu}$ (yr)	$m_{\beta\beta}$ (meV)
^{136}Xe	KamLAND-Zen [20]	$> 3.8 \times 10^{26}$	< 28 to 122
^{76}Ge	GERDA [21]	$> 1.8 \times 10^{26}$	< 79 to 180
^{130}Te	CUORE [22]	$> 2.2 \times 10^{25}$	< 90 to 305

The field has a cautionary episode. In 2001 a subset of the Heidelberg-Moscow collaboration claimed an observation in ^{76}Ge , finally quoted as $T_{1/2}^{0\nu} = 2.23_{-0.31}^{+0.44} \times 10^{25}$ yr. The Moscow part of the collaboration dissociated itself from the claim, which was criticised on statistical grounds and definitively excluded by GERDA, whose limit lies an order of magnitude above [4, 9].

4 Perspectives

4.1 Interplay with cosmology and single beta decay

Three independent observables constrain the absolute neutrino mass scale, and only their combination is informative. Cosmological surveys probe the sum $\Sigma = \sum_j m_j$ through the suppression of structure formation by massive neutrinos. Single beta decay probes m_β through the distortion of the tritium spectrum near its endpoint, the current bound being $m_\beta < 0.45$ eV at 90 % C.L. from KATRIN, obtained from 259 measurement days and essentially free of model assumptions [23]. Only $0\nu\beta\beta$ probes $m_{\beta\beta}$, and only it presupposes the Majorana nature.

The cosmological constraints have tightened sharply. Combining the first-year baryon acoustic oscillation measurements of DESI with cosmic microwave background data yields $\Sigma < 0.072$ eV at 95 % C.L., or 0.113 eV depending on the prior adopted for Σ [24]. This is to be compared with the minima imposed by oscillation data, approximately 0.059 eV for the normal ordering and 0.10 eV for the inverted one.

Taken at face value, the DESI bound lies below the inverted-ordering minimum and therefore disfavours that scenario, which is precisely the region the next generation of $0\nu\beta\beta$ experiments is designed to probe. Since the inverted band corresponds to $m_{\beta\beta}$ between roughly 15 and 50 meV, a cosmological exclusion would push the expected signal down to the normal-ordering range, where $m_{\beta\beta}$ may be of order a few meV or even vanish entirely.

This conclusion should be stated with care, since a cosmological bound is not a direct measurement. The limit depends on the prior adopted for Σ , it is partly driven by the preference of DESI data for a higher value of H_0 , and it relaxes appreciably in extended cosmological models such as dynamical dark energy, or when different Planck likelihoods and supernova datasets are combined [24]. Some dataset combinations even yield limits below the normal-ordering minimum, which would signal a tension with oscillation data rather than a determination of the ordering. The cosmological preference for the normal ordering is therefore real but model-dependent, and it does not remove the need for a direct laboratory determination.

Whatever the eventual resolution, the interplay is informative in both directions. A $0\nu\beta\beta$ signal observed while cosmology imposes a very small Σ would indicate that a mechanism other than light Majorana neutrino exchange dominates the transition.

The quantitative target follows from Table 2, a sensitivity of order 10 to 15 meV would cover the inverted band in full. The next-generation experiments, LEGEND-1000 with ^{76}Ge , nEXO with ^{136}Xe and CUPID with ^{100}Mo , are designed to reach $m_{\beta\beta}$ of order 5 to 20 meV, entering that region for the first time [20]. Whether this suffices depends critically on the value of g_A discussed in Section 2.6, the required exposure differing by more than an order of magnitude between the unquenched and strongly quenched scenarios.

Beyond the mass scale, the same heavy states that generate light Majorana masses through the seesaw of Section 1.3 can, by decaying out of equilibrium and violating CP, produce a leptonic asymmetry in the early Universe, converted into a baryonic one by sphaleron processes [25]. Leptogenesis thus links the smallness of neutrino masses, their Majorana nature and the observed matter-antimatter asymmetry, which is what gives the search its cosmological reach.

4.2 Alternative mechanisms

Observing $0\nu\beta\beta$ would not by itself identify the operator responsible. Several alternatives to light Majorana neutrino exchange exist [4].

Heavy right-handed neutrinos with $M_I \gg 100$ MeV contribute through the point-like dimension-nine operator of Section 2.3, with effective parameter $\sum_I U_{eI}^2/M_I$. Since the mixings scale as $U_{eI} \sim m_D/M_I$, the contribution is suppressed as M_I^{-3} , more steeply than the light exchange, though windows in the GeV to TeV range remain relevant and connect directly to accelerator searches.

Right-handed currents provide a second possibility. Writing the electron current as $j_\alpha^{(e)} =$

$\bar{e}\gamma_\alpha[(1 - \gamma_5) + \eta(1 + \gamma_5)]\nu_e$, the decay becomes possible even for $m(\nu) = 0$, with $\Gamma^{0\nu} \propto |\eta|^2$ [9]. In full left-right symmetric models the exchange of W_R bosons yields a dimension-nine operator suppressed by the fourth power of the new gauge boson mass.

Finally, if $U(1)_{LC}$ is broken spontaneously rather than explicitly, Goldstone's theorem requires a massless scalar, the Majoron, and the channel $(A, Z) \rightarrow (A, Z + 2) + 2e^- + \phi^0$ opens. The Majoron carries away energy, so the signature is not a peak but a continuum shifted towards higher energies relative to $2\nu\beta\beta$ [9].

Discriminating between these would require comparing half-lives across several isotopes, whose matrix element ratios differ from one mechanism to another, together with the angular distribution and individual electron energies accessible to tracking detectors, and cross-checks against accelerator searches for W_R bosons and heavy neutrinos.

5 Conclusion

Neutrino oscillations established that neutrinos are massive, but they are structurally blind to the way that mass is generated. The distinction between a Dirac and a Majorana mass term, which amounts to whether the total lepton number is conserved, therefore requires an observable sensitive to $|\Delta L| = 2$. Neutrinoless double beta decay is, with present technology, the only such process within reach.

Its amplitude rests on two conditions established in Section 2.3, the neutrino must be its own antiparticle, and a helicity flip must occur, which makes the rate proportional to the square of the effective Majorana mass $m_{\beta\beta}$ rather than merely allowed by it. Observing the decay would establish that lepton number is not conserved in nature and, through the Schechter–Valle theorem, that neutrinos are Majorana fermions whatever the underlying mechanism. The converse does not hold, since destructive interference between the Majorana phases can make $m_{\beta\beta}$ vanish even for a Majorana neutrino, and since other operators could dominate the transition.

The dominant obstacle is not experimental but theoretical. Phase-space factors are known to about 7 %, and nuclear matrix elements now agree between methods to within a factor two to three, but the effective value of the axial coupling g_A in the nuclear medium remains uncertain enough to shift the inferred limit on $m_{\beta\beta}$ by a factor four to eight, and the required exposure by more than an order of magnitude. Progress in nuclear theory is therefore as decisive for the field as progress in detector technology.

Experimentally, no signal has been found. The best limits, dominated by KamLAND-Zen with $T_{1/2}^{0\nu} > 3.8 \times 10^{26}$ yr, correspond to $m_{\beta\beta}$ below roughly 30 to 120 meV, at the upper edge of the inverted ordering band. The next generation of tonne-scale detectors aims at 10 to 15 meV, which would explore that band in full. Whether the outcome is a discovery or an exclusion, it will constrain one of the few questions in particle physics whose answer bears simultaneously on the symmetries of the Standard Model, the origin of neutrino mass, and the matter-antimatter asymmetry of the Universe.

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A Historical timeline

Year	Development
1930	Pauli postulates the neutrino
1934	Fermi formulates the theory of beta decay
1935	Goeppert-Mayer proposes $2\nu\beta\beta$, estimating $T_{1/2} > 10^{17}$ yr
1937	Majorana proposes self-conjugate fermions
1937	Racah notes the observable consequences for inverse beta decay
1939	Furry shows that Majorana neutrinos allow $0\nu\beta\beta$
1950	Inghram and Reynolds observe $2\nu\beta\beta$ geochemically in ^{130}Te
1955	Davis finds no ^{37}Cl transmutation from reactor antineutrinos
1956	Cowan and Reines detect the neutrino
1957	Two-component neutrino theory, Pontecorvo proposes oscillations
1958	Goldhaber <i>et al.</i> measure the neutrino helicity
1958	$V-A$ theory of the charged weak current
1960	Greuling and Whitten compute $\Gamma^{0\nu}$ in $V-A$ theory
1962	Maki, Nakagawa and Sakata introduce neutrino mixing
1969	Gribov and Pontecorvo, Majorana mass term for flavour neutrinos
1976	Bilenky and Pontecorvo, Dirac plus Majorana mass term
1977	Seesaw mechanism proposed
1979	Weinberg dimension-five operator
1980	Majorana CP phases identified
1981	Majoron emission channel proposed
1982	Schechter and Valle, the black box theorem
1986	Fukugita and Yanagida propose leptogenesis
1987	Elliott, Hahn and Moe, first direct $2\nu\beta\beta$ observation (^{82}Se)
1998	Super-Kamiokande, evidence for atmospheric oscillations
2001	Heidelberg-Moscow subset claims a $0\nu\beta\beta$ signal, later excluded
2001	SNO, solar neutrino flavour conversion
2015	Nobel Prize to Kajita and McDonald for oscillations
2020	GERDA final result, $T_{1/2}^{0\nu}(^{76}\text{Ge}) > 1.8 \times 10^{26}$ yr
2024	KamLAND-Zen, $T_{1/2}^{0\nu}(^{136}\text{Xe}) > 3.8 \times 10^{26}$ yr

Table 4: Main milestones in the study of the neutrino nature and of double beta decay [4, 9, 15].

B Phase-space factor

For completeness, the explicit form of the two-neutrino phase-space factor introduced in Section 2.2 is [15]

$$\begin{aligned}
 G^{2\nu} \propto \int_{m_e}^{E_0-m_e} F(Z, E_{e1}) p_{e1} E_{e1} dE_{e1} \int_{m_e}^{E_0-E_{e1}} F(Z, E_{e2}) p_{e2} E_{e2} dE_{e2} \\
 \times \int_0^{E_0-E_{e1}-E_{e2}} p_{\nu 1}^2 (E_0 - E_{e1} - E_{e2} - p_{\nu 1})^2 dp_{\nu 1}, \quad (21)
 \end{aligned}$$

where $E_0 = Q_{\beta\beta} + 2m_e$. The three integration variables correspond to the energy of each electron and the momentum of one of the two antineutrinos, the energy of the second antineutrino is fixed by overall energy conservation once the other three are specified, and is therefore not an

independent integration variable. Carrying out this integration, exactly or numerically, gives the scaling result quoted in Section 2.2, $G^{2\nu} \propto Q_{\beta\beta}^{11}$.

C Extremal values of $m_{\beta\beta}$

The effective Majorana mass of Section 2.4 is the modulus of a sum of three complex numbers, $m_{\beta\beta} = |z_1 + z_2 + z_3|$ with $|z_j| = |U_{ej}^2| m_j$, whose relative phases are unknown. The maximum is reached when the three terms are aligned, giving $m_{\beta\beta}^{\max} = \sum_j |U_{ej}^2| m_j$.

The minimum follows from a case analysis [4]. Define $r_j = |z_j| - \sum_{k \neq j} |z_k|$. At most one r_j can be positive. If some $r_j > 0$, one term is larger than the other two combined, no choice of phases can cancel it, and $m_{\beta\beta}^{\min} = r_j$. If instead $r_j \leq 0$ for every j , the three moduli satisfy the triangle inequality and can be arranged to close a triangle, so that $m_{\beta\beta}^{\min} = 0$. Combining both cases,

$$m_{\beta\beta}^{\min} = \max \left\{ 2|U_{ej}^2| m_j - m_{\beta\beta}^{\max}, 0 \right\}, \quad (22)$$

since $r_j = 2|z_j| - \sum_k |z_k|$. Experimentally, the region where $m_{\beta\beta}^{\min} = 0$ is accessible only in the normal ordering, for a lightest neutrino mass of a few meV.

D The Racah chain and helicity suppression

Shortly after Majorana's paper, Racah observed that if neutrinos are self-conjugate, the antineutrino emitted in a beta decay,

$$(A, Z) \rightarrow (A, Z + 1) + e^- + \bar{\nu}, \quad (23)$$

could in principle be absorbed by another nucleus to produce a second electron,

$$\bar{\nu} + (A, Z + 1) \rightarrow (A, Z + 2) + e^-, \quad \bar{\nu} \equiv \nu, \quad (24)$$

whereas for a Dirac neutrino only a positron could be produced [9]. This suggested a direct test of the neutrino nature using free neutrino fluxes.

The test is nonetheless inconclusive, for the reason developed in Section 2.3. The state emitted in beta decay has, to an excellent approximation, positive helicity, while the charged current can only absorb the negative-helicity component. That component has amplitude of order $m_\nu/2E$, so the cross section is suppressed by $(m_\nu/E)^2$. With $m_\nu \sim 0.05$ eV and $E \sim 1$ MeV, the suppression is of order 10^{-15} , and a null result is equally compatible with both hypotheses. Historically, Davis exposed tanks of carbon tetrachloride to the Brookhaven reactor in 1955 and set an upper limit of 2×10^{-42} cm² per atom on the antineutrino capture cross section [26]. The null result was sometimes read as evidence for a Dirac neutrino, which it is not, helicity suppression alone accounts for it.

Neutrinoless double beta decay evades this obstacle in two ways. The exchanged neutrino is virtual rather than real, propagating over a nucleon spacing inside a single nucleus rather than between two targets, and the experiment observes of order 10^{26} nuclei simultaneously over several years. The same helicity suppression applies, which is why $\Gamma^{0\nu} \propto m_{\beta\beta}^2$, but the accumulated statistics bring it within reach.

E Diagonalisation of the seesaw mass matrix

In the one-flavour case of Section 1.3, the mass matrix in the basis (ν_L, N_R) is

$$\mathcal{M} = \begin{pmatrix} 0 & m_D \\ m_D & M \end{pmatrix}, \quad (25)$$

whose eigenvalues satisfy the characteristic equation

$$\lambda^2 - M\lambda - m_D^2 = 0, \quad \lambda_{\pm} = \frac{M \pm \sqrt{M^2 + 4m_D^2}}{2}. \quad (26)$$

For $M \gg m_D$ the square root can be expanded, $\sqrt{M^2 + 4m_D^2} \simeq M (1 + 2m_D^2/M^2)$, giving

$$\lambda_+ \simeq M + \frac{m_D^2}{M} \simeq M, \quad \lambda_- \simeq -\frac{m_D^2}{M}. \quad (27)$$

The heavy eigenstate therefore has mass close to M , while the light one acquires the suppressed mass $m_\nu \simeq -m_D^2/M$ quoted in Section 1.3. The negative sign is unphysical and can be absorbed in a redefinition of the field phase, only $|m_\nu|$ being observable. The mechanism generalises to n flavours, where m_D and M become matrices and the light mass matrix reads $m_\nu \simeq -m_D^t M^{-1} m_D$.